

The 3rd *Dimension*

EULER'S RULE

One of the most brilliant mathematicians of all time was a German man called Leonhard Euler (his name is pronounced "oiler"). He wrote 75 books and was blind for the last



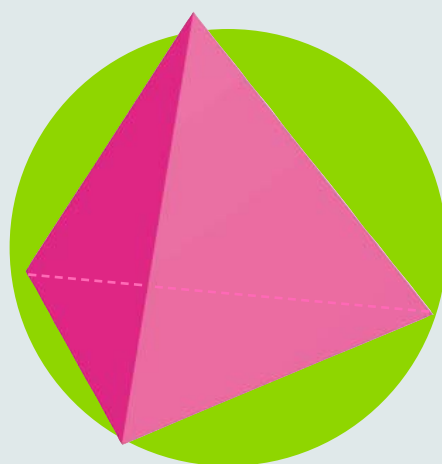
Leonhard Euler
(1707–1783)

17 years of his life, yet he did half his best work then. One of his most famous discoveries concerned

the Platonic solids. Euler found that the number of faces, edges, and vertices (corners) in a 3D shape obeys a simple mathematical rule. See if you can work it out. Fill in the table below by counting the number of faces, edges, and corners on each shape. Then look for a pattern. *Hint: for each shape, add the number of faces to the number of corners and compare the answer with the number of edges.*

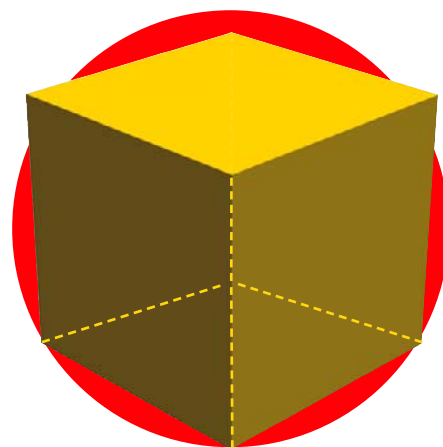
	FACES	EDGES	CORNERS
CUBE	6	12	8
TETRAHEDRON	?	?	?
OCTAHEDRON	?	?	?
DODECAHEDRON	12	30	20
ICOSAHEDRON	?	30	12

TETRAHEDRON



- The tetrahedron is made of four equilateral triangles.
- The tetrahedron is a kind of pyramid, but unlike the famous pyramids of Egypt, its base is triangular rather than square.
- The ancient Greeks thought everything was made of four elements: earth, air, fire, and water. Because the tetrahedron has the sharpest corners of the Platonic solids, the Greeks thought fire was made of tetrahedral atoms.
- Because the tetrahedron is made of triangles, it is very strong. Diamond – the hardest substance known – is strong because its atoms are arranged in a grid of connected tetrahedrons.

CUBE



- The cube is made of 6 squares joined at right angles.
- Cubes stack together more easily than any other shape. There's an infinite number of ways of stacking cubes together so they fit without any gaps.
- The Greeks chose the cube to represent the element earth, as cubes fit together so solidly. They thought rock might be made of cubic atoms.
- Some crystals, including table salt, grow naturally as cubes because the atoms within them are arranged in a cubic pattern.
- The four-dimensional version of a cube is a "hypercube". This shape can't exist in our universe, but mathematicians know it would have 32 edges, 16 corners, and 24 faces.

The ancient Greeks thought the *dodecahedron*